

# From Pixels to Patches: Pooling Strategies for Earth Embeddings

Isaac Corley, Caleb Robinson, Inbal Becker-Reshef, and Juan M. Lavista Ferres

**Abstract**—Geospatial foundation models (GFMs) increasingly publish pixel-level embedding products that practitioners reuse without access to the encoder. Downstream tasks with patch- or region-level labels must aggregate these dense pixel embeddings into a single vector, yet the choice of pooling operator is rarely examined. We show that one structural quantity—the ratio  $N/D$  of pixels per region to embedding dimension—governs whether second-order pooling (covariance, full distributional statistics) is viable. When  $N \gg D$  (e.g.  $64 \times 64$  fixed chips), second-order statistics capture intra-region heterogeneity that mean pooling discards; when  $N \approx D$  (small parcels, fields, buildings), the sample second-moment matrix is ill-conditioned and covariance collapses, while first-order distributional pooling (*mean+std*, *center-weighted*) remains stable. We benchmark 13 operators on EuroSAT (10 classes,  $N=4096$ ) and PASTIS (18 crop classes,  $\sim 85K$  parcels, median  $N=196$ ) across three GFMs (AlphaEarth, OlmoEarth-Nano, Tessera). On EuroSAT, distributional statistics shrink the random→spatial accuracy gap from 9.6 to 3.4 pp averaged across encoders (per-encoder reductions 43–76%); covariance drops from 91.0% (EuroSAT) to 58.7% F1 (PASTIS, AEF) while *mean+std* is the only training-free method to stay in the top three across the small, medium, and large parcel-size bins. We give practitioners a one-step decision rule and release *EuroSAT-Embed*, *PASTIS-Embed*, and benchmark code.<sup>1</sup>

**Index Terms**—Geospatial foundation models, embeddings, pooling, land cover classification, crop classification, spatial generalization.

## I. INTRODUCTION

A typical agricultural parcel in France covers 1.96 ha—196 pixels at 10 m resolution. A standard EuroSAT land-cover patch, by contrast, contains 4,096 pixels. Both settings require aggregating pixel embeddings from a geospatial foundation model (GFM) into a single patch-level vector, yet the number of pixels  $N$  available for that summary differs by a factor of twenty. We show this difference, relative to the embedding dimension  $D$ , determines which pooling method wins.

GFMs now publish dense pixel-level embeddings as downloadable data products [1]–[5], enabling practitioners to reuse them without access to the underlying encoder. Many downstream tasks—crop mapping [6]–[8], building detection, land-cover classification—assign a single label to a region rather

than a pixel, requiring a post-hoc aggregation step (Fig. 1). This *input-label resolution mismatch* [9] is ubiquitous: EuroSAT [10] uses  $64 \times 64$  patches, PASTIS [11] uses irregular agricultural parcels. The default aggregation—mean pooling—collapses each region to its average vector, discarding within-region variability that we show carries class signal.

Pooling is well-studied in image retrieval—generalized mean (GeM) [12], VLAD [13], NetVLAD [14], Fisher Vectors [15]—and in temporal audio modeling [16], but remains largely unexplored for GFM embeddings over irregular-polygon regions. The key challenge specific to this setting is that  $N$  varies by three orders of magnitude across region types and sizes. When  $N < D$ , the sample covariance matrix has fewer data points than parameters and is mathematically rank-deficient; even at  $N \approx D$  its eigenvalues are unstable, so second-order statistics estimated from such regions are unreliable. When  $N \gg D$ , those same statistics are well-conditioned and outperform mean pooling by a large margin. No prior work characterizes this transition or gives practitioners a decision rule.

We study post-hoc pooling across the full  $N/D$  spectrum. Using pixel embeddings from three GFMs on two benchmarks, we benchmark 11 training-free and 2 fitted pooling operators and measure how method rankings shift with region size. Our contributions:

- A practitioner-facing decision rule (Table III, Sec. V): the ratio  $N/D$  predicts when second-order operators (covariance, multi-moment stats) are viable, explaining the collapse of covariance pooling between EuroSAT ( $N/D=64$ ) and small PASTIS parcels ( $N/D \lesssim 1$ ).
- A benchmark of 11 training-free and 2 fitted operators on EuroSAT (3 GFMs, random and spatial splits, kNN and linear probes) and PASTIS, showing distributional statistics shrink the EuroSAT random→spatial accuracy gap from 9.6 pp to 3.4 pp on average (per-encoder reductions 43–76%) (Sec. V, Tables I–II).
- *EuroSAT-Embed* and *PASTIS-Embed*: pre-computed pixel-level embeddings from three GFMs, released with all benchmark code, providing the first controlled study of post-hoc pooling for GFM embedding products (Sec. II).

## II. DATA AND EMBEDDINGS

**EuroSAT.** We use the 10-class EuroSAT land-cover dataset [10], which contains 27,000 Sentinel-2 patches at  $64 \times 64$  pixels. We evaluate on both the standard random split [17] and a spatial split [18], [19] that partitions samples

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<sup>1</sup>Code: <https://github.com/isaaccorley/geopool>; data: <https://huggingface.co/datasets/isaaccorley/eurosat-embed> and <https://huggingface.co/datasets/isaaccorley/pastis-embed>.

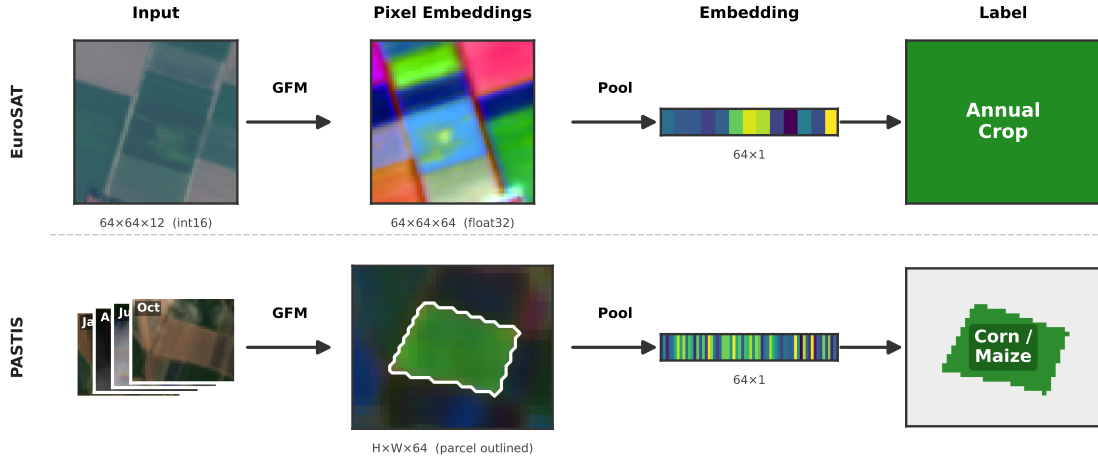


Fig. 1. **Embedding pipeline for EuroSAT (top) and PASTIS (bottom).** In both settings, a GFM encodes every pixel independently and a pooling operator aggregates those pixel vectors into one patch or parcel representation. *Top:* A  $64 \times 64$  Sentinel-2 patch is encoded pixel-wise; all 4,096 pixel embeddings are pooled into a single patch vector for land-cover classification. *Bottom:* Multi-temporal Sentinel-2 frames are encoded per-pixel; only pixels inside the parcel boundary (contour shown on a PCA pseudo-RGB visualization—first three principal components of the embedding rendered as colour) are pooled into a single parcel vector for crop-type classification. The number of pixels per parcel  $N$  varies from 5 to 3,440, making the ratio  $N/D$  a key design parameter.

by longitude to reduce train-test leakage via spatial autocorrelation.

**PASTIS.** We use the PASTIS panoptic segmentation dataset [11], which contains 2,433 Sentinel-2 time-series patches covering France with instance-level annotations of  $\sim 85,000$  agricultural parcels across 18 crop classes. Parcels are irregular polygons extracted from instance masks; parcel size varies from 5 to 3,440 pixels (median 196). We follow the official PASTIS train/test split using the folds released with the original paper (folds 1–4 train, fold 5 test). Parcels with fewer than 5 pixels are excluded.

**EuroSAT-Embed.** We construct pixel-level embedding datasets from AlphaEarth [20] (AEF, 64-d int8), OlmoEarth-Nano [21] (128-d float32,  $T=12$  subsampled time steps), and Tessera [22] (128-d float32). All three encoders ingest Sentinel-2 time series and emit a single annual per-pixel embedding; AEF and OlmoEarth additionally use patch spatial context. We apply the same three encoders to PASTIS parcels to produce *PASTIS-Embed*. We study pooling behavior, not encoder comparison.

### III. POOLING METHODS

We benchmark training-free operators that compress an  $H \times W \times D$  embedding tensor ( $N=HW$  pixels) into a single feature vector, plus two operators that fit a light additional stage on the training split.

**First-order statistics** ( $D$ -d): *mean*, *max*, generalized mean (*GeM*) [12], and *center-weighted mean* (down-weighting boundary pixels).

**Distributional statistics** ( $2D$ – $5D$ -d): *std*, *mean+std*, *mean+max*, *stats* (min/max/mean/std,  $4D$ -d), *percentiles* ( $5D$ -d), *median+IQR*, and *covariance* [23] (upper triangle,  $D(D+1)/2$ -d).

**Parametric (fit on train split):** *PCA* (mean-pooled then projected) and *Bag of Visual Words (BoVW)* [24] (mini-batch  $k$ -means histogram).

In total we evaluate 11 training-free and 2 fitted operators.

### IV. EXPERIMENTAL SETUP

We evaluate two probes: kNN ( $k=5$ , cosine distance) and multinomial logistic regression with regularization  $C$  selected by 3-fold CV on the training split. We standardize before linear probing [25]. Both probes are deterministic under our protocol; rerunning the linear probe with three seeds (0, 1, 42) yields identical accuracy ( $\sigma=0$  pp). EuroSAT results appear in Table I under random and spatial splits; PASTIS results (F1-macro) appear in Table II (OE = OlmoEarth-Nano, Tess = Tessera).

### V. RESULTS

**The  $N/D$  ratio governs second-order viability.** The ratio of pixels per region  $N$  to embedding dimension  $D$  is a proxy for the conditioning of the sample second-moment estimator: the sample covariance is mathematically rank-deficient when  $N < D$  and has unstable eigenvalues until  $N$  is several times  $D$ . This directly explains the collapse of covariance pooling between EuroSAT and small PASTIS parcels; the ranking of first-order operators (mean variants) shifts more smoothly with region size and is not predicted by  $N/D$  alone. EuroSAT patches have  $N=4,096$  and  $D \in \{64, 128\}$ , giving  $N/D \geq 32$ : second-order statistics are well-conditioned. PASTIS parcels have a median  $N=196$ ; for OlmoEarth and Tessera ( $D=128$ ) the median sits at  $N/D \approx 1.5$ , and the smallest parcels in the dataset fall below  $N=D$  entirely. Fig. 3 visualizes the transition; this single structural quantity explains every major finding below.

**When  $N \gg D$ : distributional statistics win.** On EuroSAT spatial splits, *stats* pooling reaches 91–95% accuracy (avg. 93.3%) vs. 81–92% for *mean* (avg. 86.3%), a +7.0 pp gain (Table I). *Covariance* leads on random splits for two of three encoders (OlmoEarth: 97.2%, Tessera: 97.6%). Fig. 2 shows that distributional methods are simultaneously more accurate and more robust: averaged across encoders, *mean* drops 9.6 pp from random to spatial splits while *stats* drops only 3.4 pp—a 65% reduction (per-encoder: AEF 63%, OlmoEarth 43%,

TABLE I  
EUROSAT RESULTS (ACCURACY, %): LINEAR AND KNN ( $k=5$ ) PROBES UNDER RANDOM AND SPATIAL SPLITS. BEST PER COLUMN IN **BOLD**, SECOND-BEST IN *italics*. DIM = OUTPUT EMBEDDING SIZE RELATIVE TO ENCODER DIMENSION  $D$ . **STATS** HIGHLIGHTED AS THE RECOMMENDED METHOD IN THE  $N \gg D$  REGIME. \*OLMOEARTH-NANO; OE = OLMOEARTH; TESS = TESSERA.

| Pooling                                | Dim        | Linear probe  |             |             |             |              |             |             |             | kNN probe     |             |             |             |              |             |             |             |
|----------------------------------------|------------|---------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
|                                        |            | Spatial split |             |             |             | Random split |             |             |             | Spatial split |             |             |             | Random split |             |             |             |
|                                        |            | AEF           | OE*         | Tess        | Avg         | AEF          | OE*         | Tess        | Avg         | AEF           | OE*         | Tess        | Avg         | AEF          | OE*         | Tess        | Avg         |
| Mean                                   | $D$        | 81.1          | 91.8        | 86.0        | 86.3        | 95.3         | 96.4        | 96.1        | 95.9        | 85.5          | 87.8        | 84.7        | 86.0        | 94.6         | 94.8        | 94.9        | 94.8        |
| Std                                    | $D$        | 77.1          | 90.5        | 85.4        | 84.3        | 90.9         | 94.8        | 95.3        | 93.7        | 78.1          | 81.7        | 84.0        | 81.3        | 93.7         | 93.3        | 94.1        | 93.7        |
| GeM                                    | $D$        | 83.2          | 91.7        | 87.3        | 87.4        | 95.5         | 95.9        | 96.4        | 95.9        | 85.7          | 86.6        | 85.4        | 85.9        | 94.6         | 93.3        | 95.1        | 94.3        |
| Center-Weighted                        | $D$        | 85.7          | 91.9        | 87.6        | 88.4        | 96.1         | 96.8        | 96.1        | 96.3        | 86.0          | 87.8        | 85.5        | 86.4        | <b>95.4</b>  | <b>95.0</b> | 95.5        | <b>95.3</b> |
| Max                                    | $D$        | 86.1          | 90.6        | 89.8        | 88.8        | 92.9         | 92.9        | 93.7        | 93.2        | 80.5          | 85.5        | 80.8        | 82.3        | 90.5         | 91.4        | 92.5        | 91.5        |
| Mean+Std                               | $2D$       | 84.2          | 93.4        | 94.8        | 90.8        | 96.1         | 96.6        | 97.1        | 96.6        | 86.4          | 88.3        | 86.7        | 87.1        | 94.9         | 94.9        | 95.3        | 95.0        |
| Mean+Max                               | $2D$       | 89.1          | 93.3        | 94.4        | 92.3        | 96.1         | 96.3        | 96.5        | 96.3        | 85.3          | 87.9        | 86.9        | 86.7        | 94.7         | 94.4        | <b>95.8</b> | 94.9        |
| Median+IQR                             | $2D$       | 81.7          | 93.2        | 88.4        | 87.7        | 94.8         | 95.9        | 95.9        | 95.5        | 86.0          | 86.0        | 85.7        | 85.9        | 94.1         | 94.0        | 94.4        | 94.1        |
| Percentiles                            | $5D$       | 85.6          | 93.8        | 91.2        | 90.2        | 95.9         | 97.0        | 97.0        | 96.6        | 86.7          | 87.5        | 86.2        | 86.8        | 94.6         | 94.9        | 95.4        | 95.0        |
| Stats                                  | $4D$       | <b>91.0</b>   | <b>94.0</b> | <b>94.9</b> | <b>93.3</b> | <b>96.2</b>  | 96.6        | <b>97.3</b> | <b>96.7</b> | <b>86.9</b>   | 87.6        | <b>89.3</b> | <b>87.9</b> | 95.0         | 94.4        | 95.6        | 95.0        |
| Covariance                             | $D(D+1)/2$ | 89.1          | 93.4        | 93.7        | 92.1        | 95.8         | <b>97.2</b> | <b>97.6</b> | <b>96.8</b> | 84.4          | 84.8        | <b>89.8</b> | 86.4        | 93.4         | 94.3        | 94.5        | 94.1        |
| <i>Parametric (fit on train split)</i> |            |               |             |             |             |              |             |             |             |               |             |             |             |              |             |             |             |
| PCA                                    | 64         | 79.7          | 91.6        | 80.8        | 84.0        | 95.2         | 95.6        | 95.0        | 95.3        | 85.3          | <b>88.4</b> | 86.1        | 86.6        | 93.5         | 94.1        | 92.7        | 93.5        |
| BoVW                                   | 128        | 88.9          | 88.2        | 88.7        | 88.6        | 94.4         | 94.5        | 96.3        | 95.1        | 84.0          | 82.1        | 87.4        | 84.5        | 90.7         | 90.3        | 90.2        | 90.4        |

TABLE II  
PASTIS RESULTS (F1-MACRO, %): LINEAR AND KNN ( $k=5$ ) PROBES PER ENCODER. BEST PER COLUMN IN **BOLD**, SECOND-BEST IN *italics*. **MEAN+STD** HIGHLIGHTED AS THE RECOMMENDED METHOD FOR VARIABLE-SIZE REGIONS. OE = OLMOEARTH-NANO; TESS = TESSERA.

| Pooling                                | Linear probe |             |             |             | kNN probe   |             |             |             |
|----------------------------------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                        | AEF          | OE          | Tess        | Avg         | AEF         | OE          | Tess        | Avg         |
| Mean                                   | 64.5         | 54.7        | 49.0        | 56.1        | 56.5        | <b>41.1</b> | <b>43.8</b> | <b>47.1</b> |
| Std                                    | 49.0         | 22.8        | 40.3        | 37.4        | 48.2        | 14.4        | 37.2        | 33.3        |
| GeM                                    | 62.8         | 49.7        | 48.1        | 53.5        | 54.2        | 36.6        | 43.3        | 44.7        |
| Center-Weighted                        | <i>65.3</i>  | <i>54.8</i> | 48.9        | <i>56.3</i> | <b>57.1</b> | <i>40.3</i> | <i>43.7</i> | <i>47.0</i> |
| Max                                    | 57.4         | 38.8        | 45.5        | 47.2        | 46.0        | 27.6        | 39.3        | 37.6        |
| Mean+Std                               | <b>65.5</b>  | <b>54.9</b> | 48.7        | <b>56.4</b> | 56.7        | 39.7        | 43.5        | 46.6        |
| Mean+Max                               | 65.0         | 54.4        | 48.8        | 56.1        | 53.0        | 35.2        | 42.7        | 43.6        |
| Median+IQR                             | 64.3         | 50.8        | 48.4        | 54.5        | 55.7        | 35.7        | 43.3        | 44.9        |
| Percentiles                            | 65.3         | 52.4        | <b>49.4</b> | 55.7        | 56.6        | 39.1        | 43.0        | 46.3        |
| Stats                                  | 65.0         | 54.0        | 48.4        | 55.8        | 52.1        | 34.3        | 42.5        | 43.0        |
| Covariance                             | 58.7         | 42.4        | 47.0        | 49.4        | 47.6        | 22.4        | 39.6        | 36.6        |
| <i>Parametric (fit on train split)</i> |              |             |             |             |             |             |             |             |
| PCA                                    | 64.4         | 47.9        | 48.2        | 53.5        | 56.2        | 38.8        | 43.3        | 46.1        |
| BoVW                                   | 44.7         | 23.0        | 15.5        | 27.8        | 43.6        | 22.1        | 14.1        | 26.6        |

Tessera 76%). *Mean+max* (4.0 pp) and *covariance* (4.8 pp) also exhibit small gaps. kNN trends are consistent, with *stats* leading on spatial splits (avg. 87.9%) and *center-weighted mean* on random splits (avg. 95.3%).

**When  $N \approx D$ : second-order statistics collapse.** On PASTIS (Table II), *covariance* drops from 91.0% on EuroSAT to 58.7% F1 on AEF—a 32 pp collapse—because many parcels have  $N < D$ , making the sample covariance matrix rank-deficient. *Stats* similarly underperforms. In contrast, *mean+std* (avg. 56.4%, best on OlmoEarth at 54.9%, second-best on AEF at 65.5%) and *center-weighted mean* (avg. 56.3%) remain robust by preserving first-order distributional structure without requiring stable second moments. First-order methods (*mean*, *center-weighted*) are now competitive, unlike on EuroSAT. Tessera’s lower absolute scores across

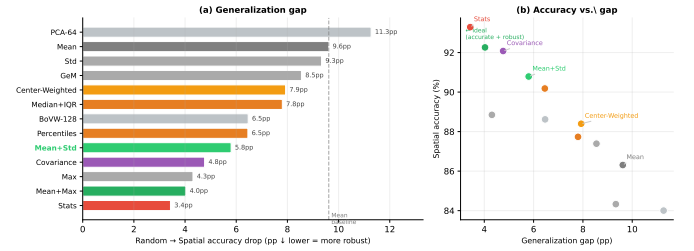


Fig. 2. **Distributional pooling is more accurate and more robust under spatial shift.** (a) Accuracy drop when moving from a random to a geographically disjoint split, averaged across three GFMs—shorter bars are better. Mean pooling drops 9.6 pp; stats drops only 3.4 pp, a 65% reduction. (b) Spatial accuracy vs. generalization gap per method. Methods in the upper-left are both accurate and robust; distributional methods cluster there while mean pooling sits lower-right.

all methods (best: 49.4%) reflect encoder-level differences in how temporal information is distilled; the pooling-method *rankings*, however, follow the same  $N/D$  trend as the other two encoders.

**Region-size breakdown.** Fig. 4 ranks methods within four parcel-size bins averaged across all three embedding sources. For tiny parcels ( $N < 50$ ), *center-weighted mean* leads (avg. rank 2.0), followed by *mean+max* (3.0) and *stats* (4.0). For small parcels (50–150 px) *center-weighted mean* again ranks first (2.0). For medium (150–500 px) parcels *mean+std* ranks first (2.67); for large ( $> 500$  px) parcels *mean+max* ranks first (1.67), with *mean+std* third (3.67). *Covariance* sits near the bottom for tiny and small bins (avg. rank  $> 11$ ) and recovers only to mid-pack (rank 8) in the large bin—it never leads at the parcel scale, consistent with  $N/D$  remaining  $\lesssim 1$ –5 even for the largest PASTIS parcels at  $D=128$ . *Mean+std* is the only training-free method that stays in the top-3 for the small, medium, and large bins, making it the practical default for variable-size regions.

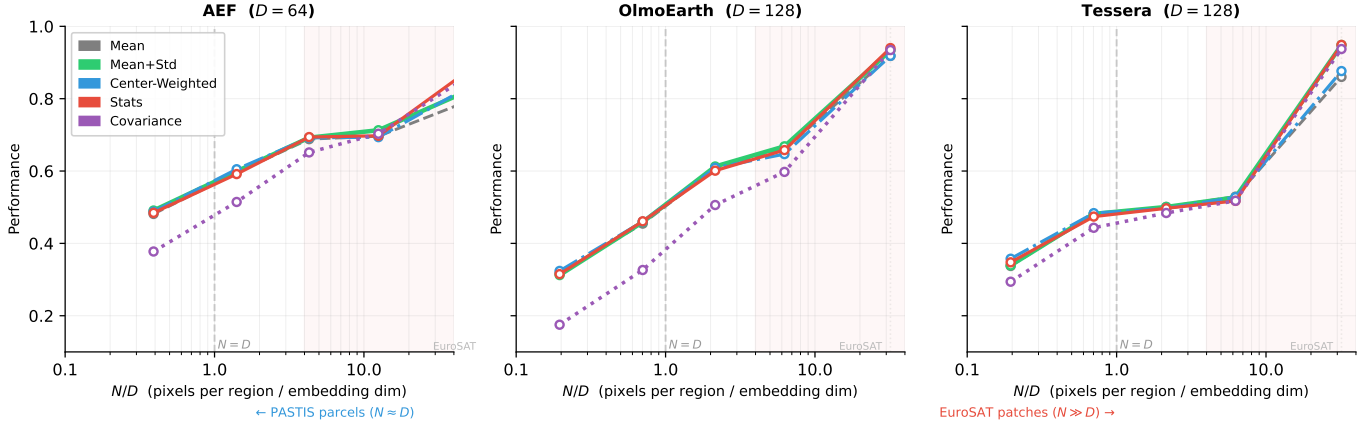


Fig. 3. **Pooling performance as a function of  $N/D$ , the ratio of pixels per region to embedding dimension.** Each panel shows per-bin F1 (PASTIS, four parcel-size bins) and accuracy (EuroSAT) plotted against  $N/D$  on a log scale for one GFM. The dashed vertical line marks  $N=D$ . To the left, where parcels are small and  $N \approx D$ , covariance (purple) collapses because its sample estimate becomes rank-deficient, while mean+std remains strong. Moving right toward EuroSAT patches where  $N \gg D$ , covariance recovers and leads. Mean+std stays competitive across the full range.

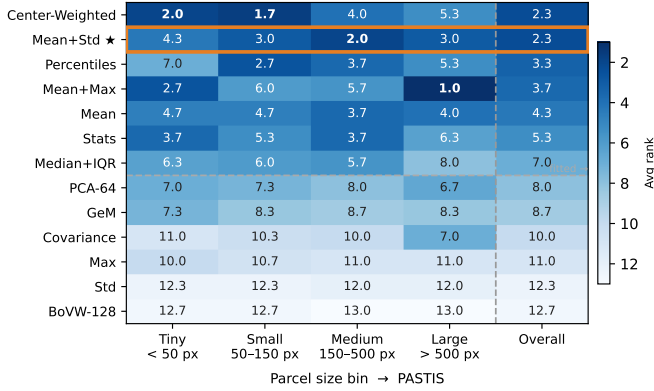


Fig. 4. **Method rankings shift as parcel size grows.** Each cell shows the average rank across AEF, OlmoEarth, and Tessera within each parcel-size bin—darker blue is a better rank (1=best). Center-weighted mean and mean+max dominate the tiny bin; mean+std rises as parcels grow; covariance moves from near-bottom (tiny/small) to mid-pack (large) but never leads at the parcel scale, since  $N/D$  remains modest. Mean+std (★, highlighted in orange) is the only training-free method that stays in the top 3 for the small, medium, and large bins. The dashed vertical line separates per-bin results from the overall ranking.

#### A. PASTIS Crop Classification

Fig. 4 summarizes the per-bin ranking. AEF achieves the highest overall F1 because its  $D=64$  gives a more favorable  $N/D$  ratio at small parcel sizes than OlmoEarth or Tessera ( $D=128$ ). This is consistent with the  $N/D$  prediction: AEF’s smaller embedding space makes second-order statistics viable at smaller parcel sizes.

## VI. DISCUSSION

**Recommendation.** Table III gives an empirical decision rule indexed by  $N/D$ :  $N/D$  formally predicts when second-order pooling is viable, and the small/large region rows are empirical defaults supported by our two benchmarks. When  $N \gg D$  (large fixed-size patches, e.g.  $64 \times 64$  at 10 m), use *stats* or *covariance* for peak accuracy; *stats* is more robust to

TABLE III

PRACTITIONER CHEAT-SHEET: PICK A POOLING OPERATOR FROM  $N$  (PIXELS PER REGION) AND  $D$  (EMBEDDING DIMENSION). APPLY AT INFERENCE TIME, NO TRAINING.

| Regime                                   | Recommended                               | Fallback               |
|------------------------------------------|-------------------------------------------|------------------------|
| $N \gg D$ (fixed chips, $\gtrsim 1k$ px) | <i>stats</i> (or <i>covariance</i> )      | <i>mean+std</i>        |
| $N \approx D$ or variable-size regions   | <i>mean+std</i>                           | <i>center-weighted</i> |
| Tiny ( $N < D$ )                         | <i>center-weighted</i><br><i>mean+max</i> | or <i>mean+std</i>     |

distribution shift. When  $N \approx D$  or regions are variable-size (parcels, fields, buildings), use *mean+std*: it wins the PASTIS linear-probe average (56.4% F1) and is the only training-free method that stays in the top-3 across the small, medium, and large parcel-size bins. For predominantly tiny regions ( $N < D$ ), *center-weighted mean* dominates. *Mean* remains a useful lower bound. Accuracy and robustness move together here: *stats* and *mean+std* are simultaneously more accurate and more robust than *mean*, with no observed accuracy–robustness tradeoff.

**Why distributional statistics help.** Mean pooling collapses each region to its average vector, discarding intra-region heterogeneity. A residential neighborhood and an industrial zone may share a mean embedding while differing in variance: the mean erases that distinction; *stats* and *mean+std* preserve it. This signal is reliable only when  $N$  is large enough to estimate higher moments—which is precisely what  $N/D$  measures.

**Practical applications.** The decision rule maps directly onto GIS zonal-statistics workflows [26], [27], where raster values are aggregated over vector polygons (agricultural fields, building footprints, administrative units, ecological zones). As GFM embedding products become available at continental scale [1], practitioners running polygon-level inference—crop monitoring [6], habitat classification, urban mapping—can apply Table III at inference time with no additional training.

**Related work.** GFM embedding products have proliferated [2]–[4], [20]–[22], but post-hoc pooling for these embed-

dings is largely unstudied; the closest prior comparisons are in image retrieval (GeM [12], VLAD [13], NetVLAD [14], Fisher vectors [15]), bilinear/covariance pooling [23], and aggregation in temporal audio [16]. Our setting differs in that  $N$  varies by three orders of magnitude across regions, which is what makes the  $N/D$  ratio actionable.

**Limitations.** We evaluate on EuroSAT (10 classes) and PASTIS (18 crop classes), both Sentinel-2 at 10 m, with three embedding sources; broader coverage (e.g., BigEarthNet [28], fMoW [29], multi-label or dense-prediction settings) would strengthen conclusions. OlmoEarth is evaluated with  $T=12$  subsampled time steps; AEF and Tessera use the encoders' default annual representation. The temporal aggregation strategy per encoder is fixed rather than ablated. We study only patch- and parcel-level labels, not semantic segmentation. We focus on training-free pooling; learned aggregation [30] may improve further but requires encoder access.

## VII. CONCLUSION

The pixel-to-dimension ratio  $N/D$  is a principled, single-number criterion for whether second-order pooling of GFM embeddings is viable, anchoring an otherwise empirical operator choice. We give a one-step recommendation in Table III that practitioners can apply at inference time, and release *EuroSAT-Embed*, *PASTIS-Embed*, and benchmark code to reproduce every result.

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